

Logistic Regression (Loan Approval Prediction)

Data Loading & Understanding

QUESTION 1

- Load the dataset into Python.
- Display the first 5 rows.
- Check the shape of the dataset.
- Display the data types of each column.

ANSWER

```
[9]: df = pd.read_csv('loan_approval - loan_approval.csv')
df.head(), df.shape, df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2000 entries, 0 to 1999
Data columns (total 8 columns):
#   Column          Non-Null Count  Dtype  
---  -
0    name            2000 non-null  object  
1    city            2000 non-null  object  
2    income          2000 non-null  int64   
3    credit_score    2000 non-null  int64   
4    loan_amount     2000 non-null  int64   
5    years_employed  2000 non-null  int64   
6    points          2000 non-null  int64   
7    loan_approved   2000 non-null  bool    
dtypes: bool(1), int64(5), object(2)
memory usage: 111.5+ KB

[9]: (   name            city  income  credit_score  loan_amount \
0  Allison Hill      East Jill  113810         389        39698
1  Brandon Hall    New Jamesside  44592         729        15446
2  Rhonda Smith    Lake Roberto  33278         584        11189
3  Gabrielle Davis West Melanieview  127196        344        48823
4  Valerie Gray     Mariastad   66048         496        47174

   years_employed  points  loan_approved
0             27      50           False
1             28      55           False
2             13      45           False
3             29      50           False
4              4      25           False ,
(2000, 8),
None)
```

QUESTION 2

Data Cleaning

ANSWER

- Check for missing values in the dataset.
- Handle missing values appropriately.
- Identify categorical columns present in the dataset.

```
None)
[15]: df.isnull().sum()

[15]: name          0
      city          0
      income        0
      credit_score  0
      loan_amount   0
      years_employed 0
      points         0
      loan_approved  0
      dtype: int64

[18]: df.dtypes

[18]: name          object
      city          object
      income        int64
      credit_score  int64
      loan_amount   int64
      years_employed int64
      points         int64
      loan_approved bool
      dtype: object
```

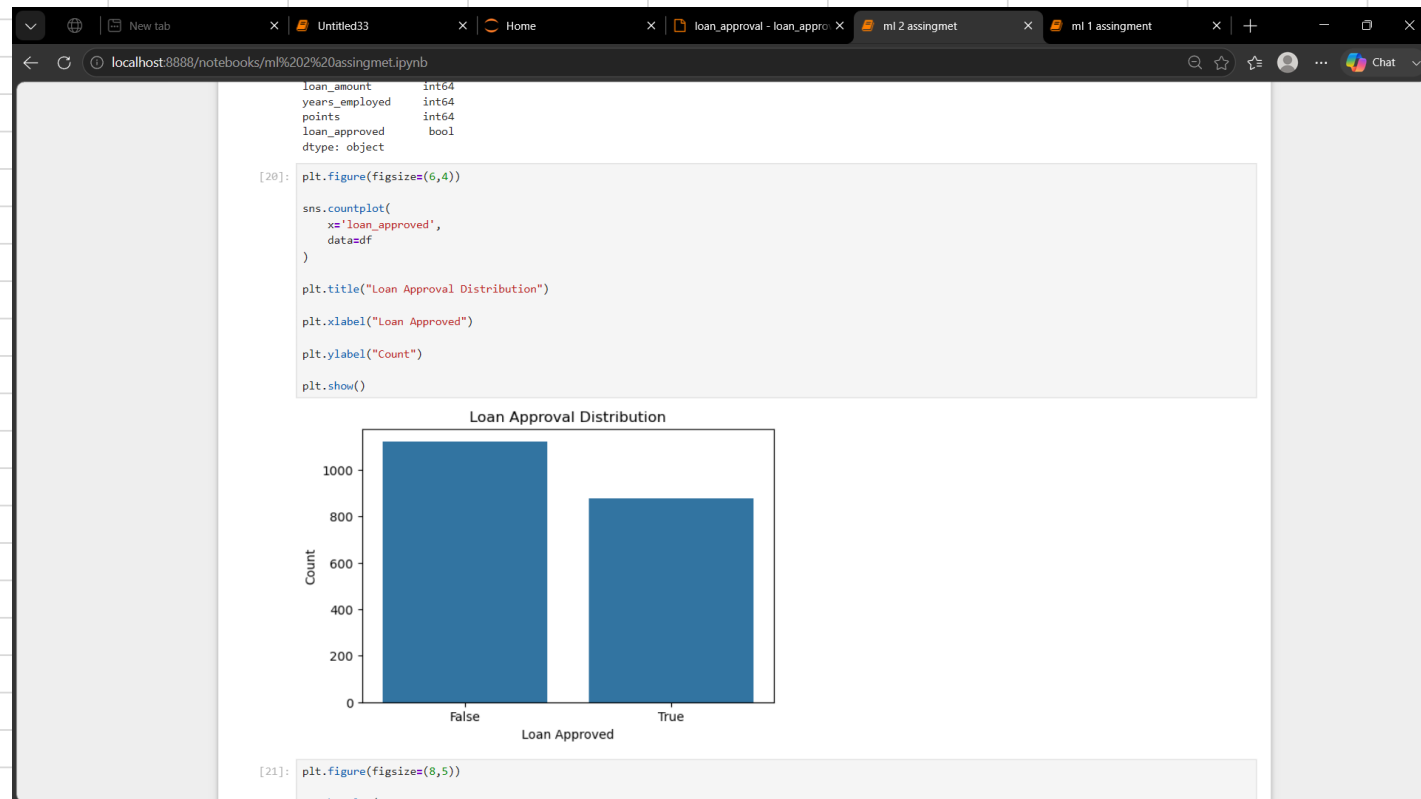
There is no null values in dataset

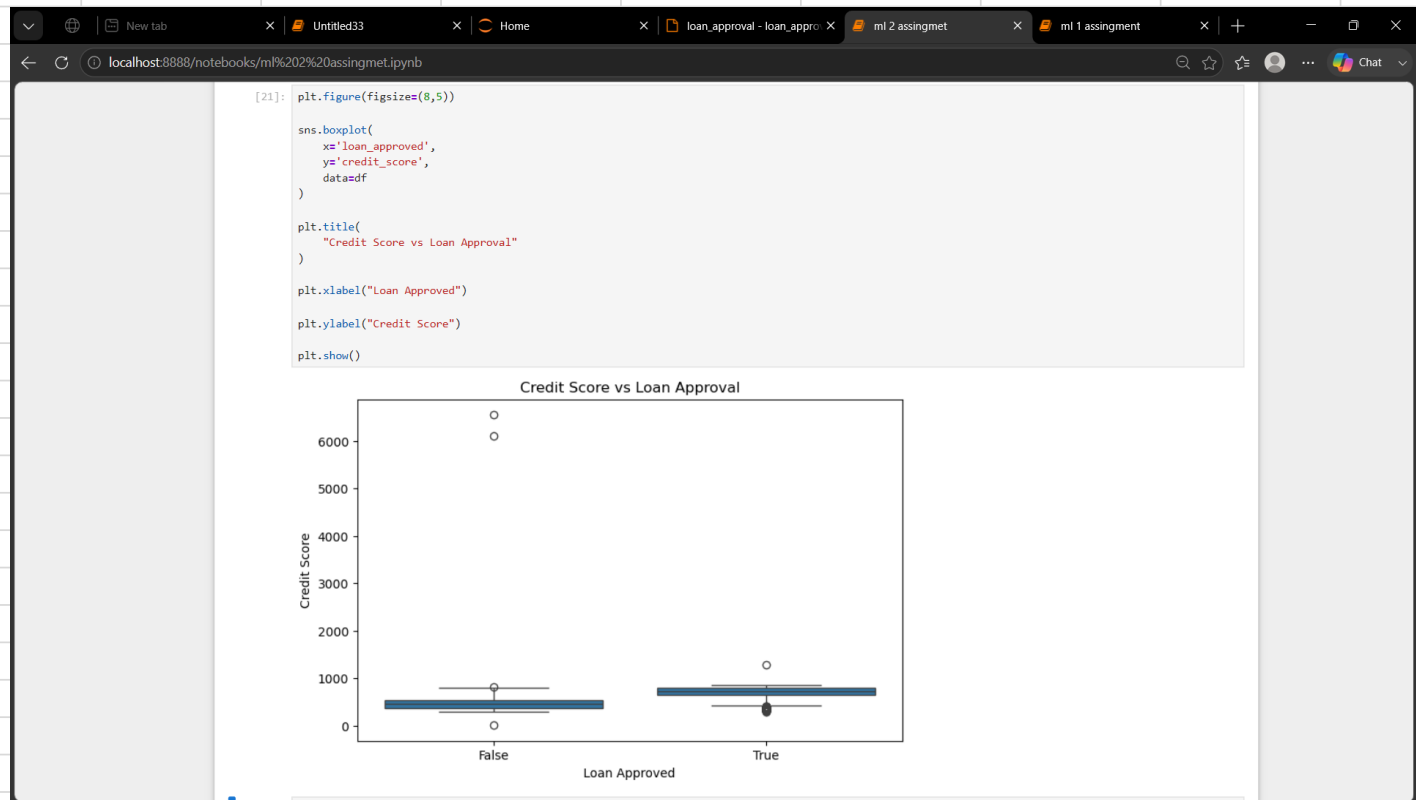
QUESTION 3

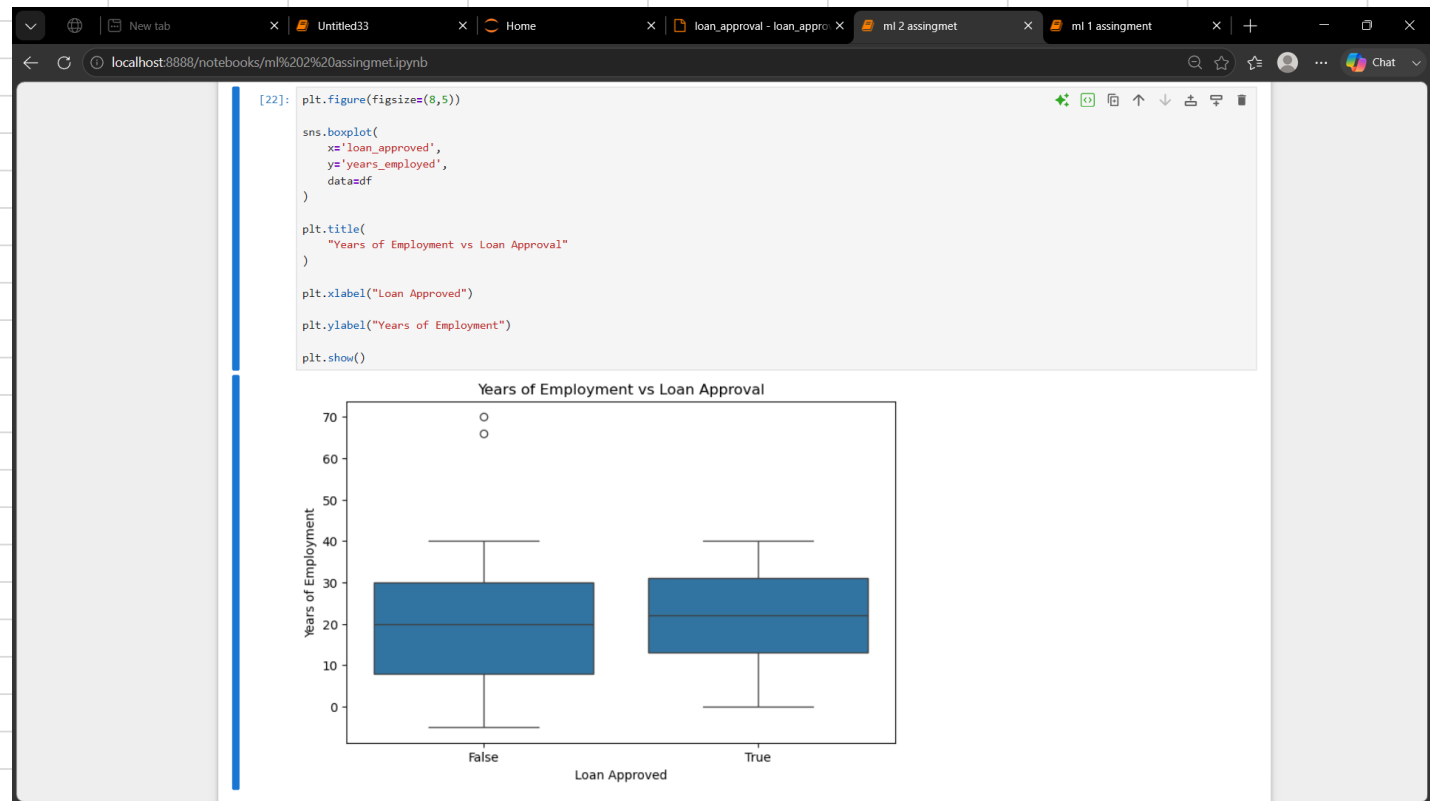
Exploratory Data Analysis (EDA)

ANSWER

- a) Plot the distribution of the target variable.
- b) Analyze the relationship between one numerical feature and the target variable.
- c) Analyze the relationship between Years of Employment and the target variable.





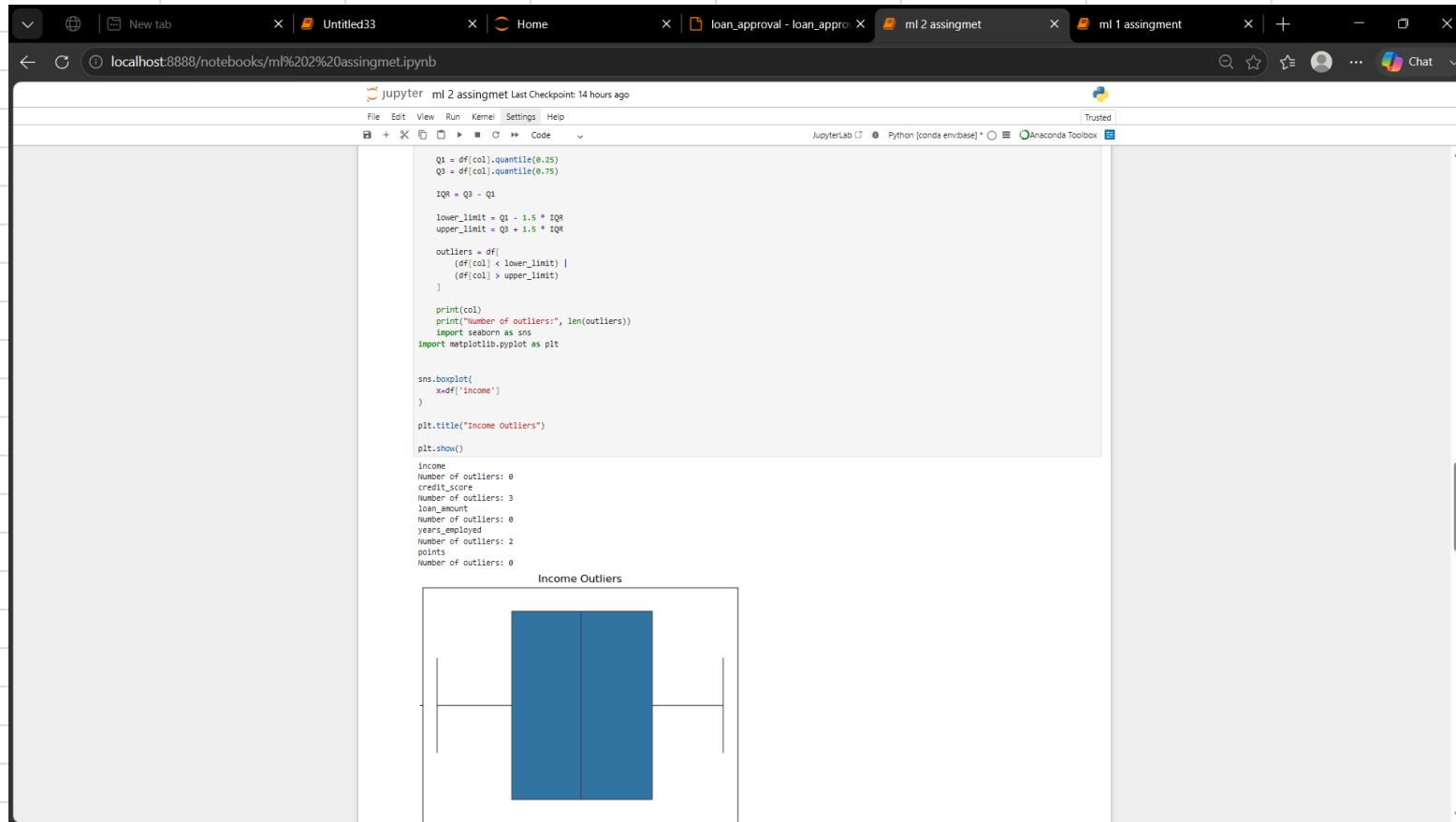


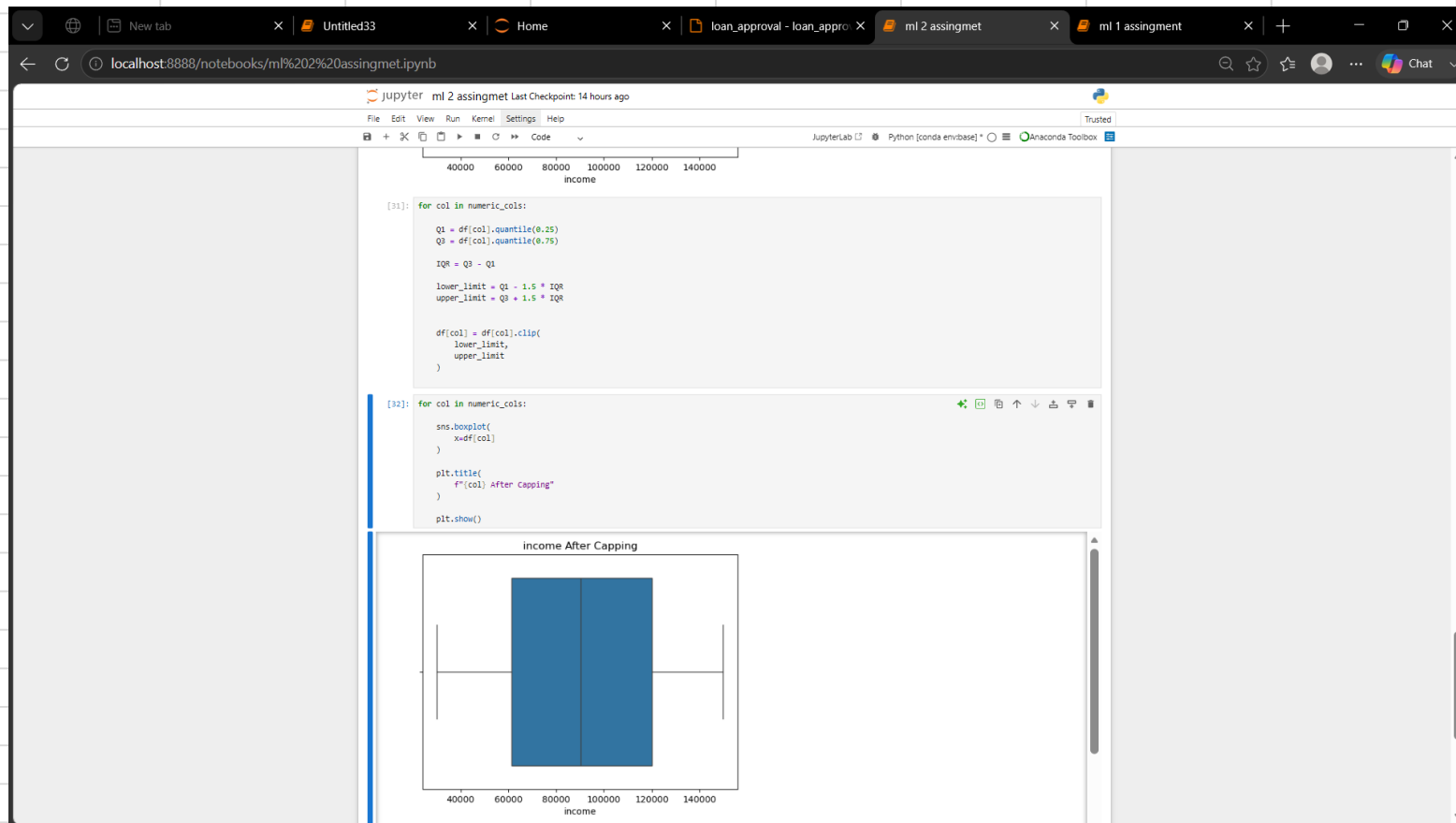
QUESTION 4

Outlier Detection and Treatment

- a) Detect outliers in numerical columns using the IQR method.
- b) Treat outliers using capping techniques

ANSWER

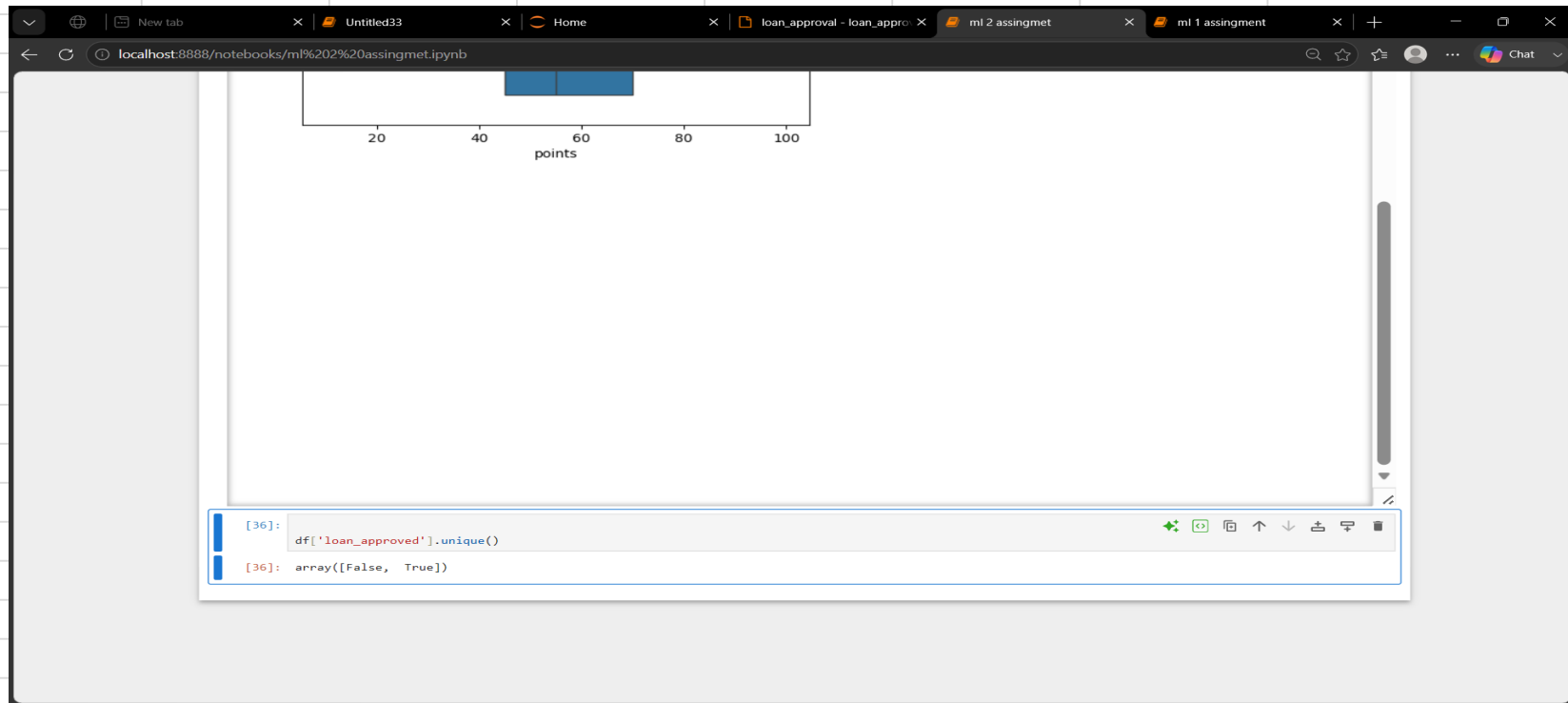




QUESTION 5

Convert target variables into numerical format suitable for model building and drop unnecessary columns.

ANSWER



AFTER


```
localhost:8888/notebooks/ml%202%20assingmet.ipynb
36]: array([False,  True])

39]: df['loan_approved'] = df['loan_approved'].astype(int)
df['loan_approved'].head(10)

39]: 0      0
      1      0
      2      0
      3      0
      4      0
      ..
     95      1
     96      1
     97      1
     98      0
     99      1
      Name: loan_approved, Length: 100, dtype: int64

[ ]: 
[ ]:
```

QUESTION 6

Feature Selection and Data Splitting

- a) Separate independent variables (X) and dependent variable (y).
- b) Split the dataset into training and testing sets.

ANSWER

```
99      1
      Name: loan_approved, Length: 100, dtype: int64

[45]: X = df.drop('loan_approved', axis=1)
      y = df['loan_approved']
      X.head(10)
      print(X.shape)
      print(y.shape)

      (2000, 7)
      (2000,)
```

localhost:8888/notebooks/ml%202%20assingmet.ipynb

(2000,)

[48]: X_train, X_test, y_train, y_test = train_test_split(
X,
y,
test_size=0.2,
random_state=42
)
print(X_train.shape)
print(X_test.shape)

print(y_train.shape)
print(y_test.shape)

(1600, 7)
(400, 7)
(1600,)
(400,)

[]:

[]:

QUESTION 7		Apply feature scaling to the dataset using StandardScaler.			
ANSWER		soory			

QUESTION 8**Logistic Regression Model Building**

- a) Train a Logistic Regression model.
b) Predict the output for test data.

ANSWER

```
localhost:8888/notebooks/Untitled35.ipynb?

(2000, 8),
None)

[35]: from sklearn.linear_model import LogisticRegression
      from sklearn.model_selection import train_test_split
      from sklearn.metrics import accuracy_score, classification_report
      X = df.drop('loan_approved', axis=1)
      y = df['loan_approved']
      X_train, X_test, y_train, y_test = train_test_split(
          X, y,
          test_size=0.2,
          random_state=42
      )
      log_model = LogisticRegression()
      log_model.fit(X_train, y_train)

      print("Model Training Completed")

-----
ValueError                                Traceback (most recent call last)
~\AppData\Local\Temp\ipykernel_17248\1434442466.py in ?()
      8     test_size=0.2,
      9     random_state=42
     10 )
     11 log_model = LogisticRegression()
--> 12 log_model.fit(X_train, y_train)
```

QUESTION 9		Model Evaluation – Confusion Matrix		
		a) Generate the confusion matrix.		
ANSWER		b) Interpret the results.		

QUESTION 10		Model Evaluation – ROC Curve & AUC		
		a) Plot the ROC curve.		
ANSWER		b) Calculate the AUC score.		